

Data Acquisition and Survey Methods

Group 10

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Exercise habits and their impact on students' focus

Group 10 investigates whether students' physical exercise habits are related to their reported level of mental brain fog during lectures. The original proposal referred to exercise frequency and academic confidence. In the final survey data, the available Group 10 variables are exercise minutes, preferred exercise type, and brain fog. Therefore, the analysis below focuses on these measured variables.

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Load data

```
library(tidyverse)
library(kableExtra)

# Full dataset
total_survey <- read_csv("Survey_results_2026.csv", show_col_types = FALSE)

# Only Group 10 data
survey <- total_survey %>%
  select(
    starts_with("DQ"),
    starts_with("G10")
  )

# Remove strings
names(survey) <- names(survey) %>%
  str_remove("^DQ\\s*") %>%
  str_remove("^G10\\s*")

# Rename columns
survey <- survey %>%
  rename(
    age = 1,
    gender = 2,
    academic_program = 3,
    exercise_minutes = 4,
    exercise_type = 5,
    brain_fog = 6
  )

# Factorize
survey <- survey %>%
  mutate(
    age = factor(age, levels = c("18-20", "21-23", "23-25", "25-30", "30-35", "over 35"), ordered = TRUE),
    exercise_type = as.factor(exercise_type),
    gender = as.factor(gender),
    academic_program = as.factor(academic_program)
  )
```

Survey questions and variables

The analysis is based on the mandatory demographic questions and the three Group 10 survey questions.

Variable	Survey question	Type
age	How old are you?	Quantitative
gender	What is your gender?	Categorical
academic_program	What is your academic program?	Categorical
exercise_minutes	How many minutes of physical exercise did you complete in the last 7 days?	Quantitative

Variable	Survey question	Type
<code>exercise_type</code>	What is your preferred type of exercise?	Categorical
<code>brain_fog</code>	Rate your average level of mental brain fog during lectures on a scale of 1 = none to 5 = severe.	Ordinal / quantitative scale

Research questions and hypotheses

RQ1: Do students who complete more minutes of physical exercise report lower levels of brain fog during lectures?

Hypothesis: Exercise minutes and brain fog are negatively associated.

RQ2: Does reported brain fog differ by preferred type of exercise?

Hypothesis: Brain fog levels differ between exercise-type groups.

RQ3: Is age negatively associated with the number of minutes spent on physical exercise per week?

Hypothesis: Older students report fewer exercise minutes.

Exploratory data analysis

The exploratory analysis uses one visualization for each research question. The plots include both quantitative and categorical variables and help identify patterns before formal hypothesis testing.

RQ1: Exercise minutes and brain fog

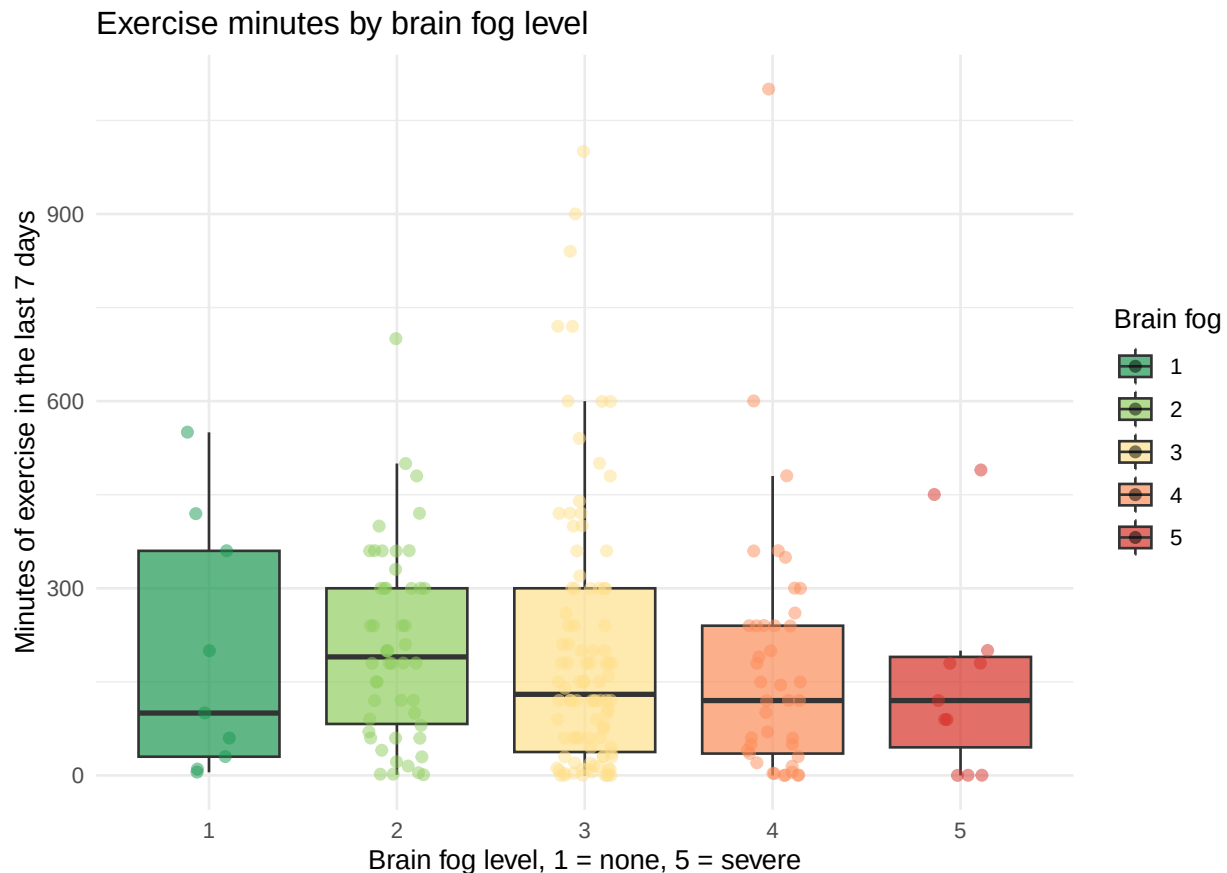
We built a layered data visualization using the `ggplot2` package to analyze how weekly exercise minutes vary across different levels of brain fog. We started by initializing the plot with the `ggplot` function, mapping the `survey` dataset. Inside the global aesthetics, we converted the `brain_fog` variable into a discrete factor so R would treat it as a categorical x-axis with five distinct groups, placed `exercise_minutes` on the continuous y-axis, and set the box colors to change dynamically based on the brain fog level.

To visualize the distribution, we overlaid two distinct geometric layers. The first layer uses `geom_boxplot` to construct the standard boxplots showing the median, quartiles, and whiskers for each category. We set the transparency to 0.7 and explicitly hid the default outliers using `outlier.shape = NA` because we planned to show all raw data points instead. Directly on top of this, we added a `geom_jitter` layer to plot every individual response as a dot. By setting a jitter width of 0.15, we spread the dots out slightly horizontally so they wouldn't stack invisibly on top of each other, and a 0.5 transparency allowed us to see data density through overlapping points.

Finally, we customized the appearance and context of the chart. We applied a custom color palette using `scale_fill_manual` to dye the boxes and named the legend "Brain fog", while using `scale_color_manual` with the guide set to none to ensure the jittered dots matched the box colors without spawning a redundant second legend. We then used the `labs` function to give the plot a clear title and explicitly define our axes, ensuring the reader knows the y-axis tracks the last seven days of exercise and the x-axis spans from none to severe. The entire visual is wrapped in `theme_minimal()`, which strips away the default heavy grey grid lines to give the final plot a clean, professional aesthetic. The color scale is green for low brain fog and red for severe brain fog.

```
ggplot(survey, aes(x = factor(brain_fog), y = exercise_minutes, fill = factor(brain_fog))) +
  geom_boxplot(alpha = 0.7, outlier.shape = NA) +
  geom_jitter(aes(color = factor(brain_fog)), width = 0.15, alpha = 0.5, size = 1.8) +
```

```
scale_fill_manual(values = brain_fog_cols, name = "Brain fog") +
scale_color_manual(values = brain_fog_cols, guide = "none") +
labs(
  title = "Exercise minutes by brain fog level",
  x = "Brain fog level, 1 = none, 5 = severe",
  y = "Minutes of exercise in the last 7 days"
) +
theme_minimal()
```



Finding:

To investigate our research question, we visualized the relationship between weekly physical activity and academic performance by plotting the quantitative variable of weekly exercise minutes against the ordinal categorical variable of reported brain fog levels using a combined boxplot and jittered scatter plot. Our initial hypothesis stated that exercise minutes and brain fog are negatively associated, meaning that higher levels of physical activity would correspond to lower levels of cognitive fatigue.

The empirical data, however, reveals a more nuanced, non linear relationship rather than a strict, steady downward trend. The central tendencies, indicated by the median lines, actually peak at a mild brain fog level of 2 (around 200 minutes) and show only a gradual decline through levels 3 and 4, while students reporting zero brain fog at level 1 display a surprisingly low median of approximately 100 minutes. Furthermore, the high variability and vertical spread within levels 1 through 3 including several highly active outliers in level 3 exercising up to 1,000 minutes demonstrate that moderate brain fog is common across a wide range of activity levels.

Nevertheless, our hypothesis holds true at the ultimate extreme of the spectrum. The distribution for level 5 is highly compressed and strictly bounded, showing that students experiencing severe brain fog almost

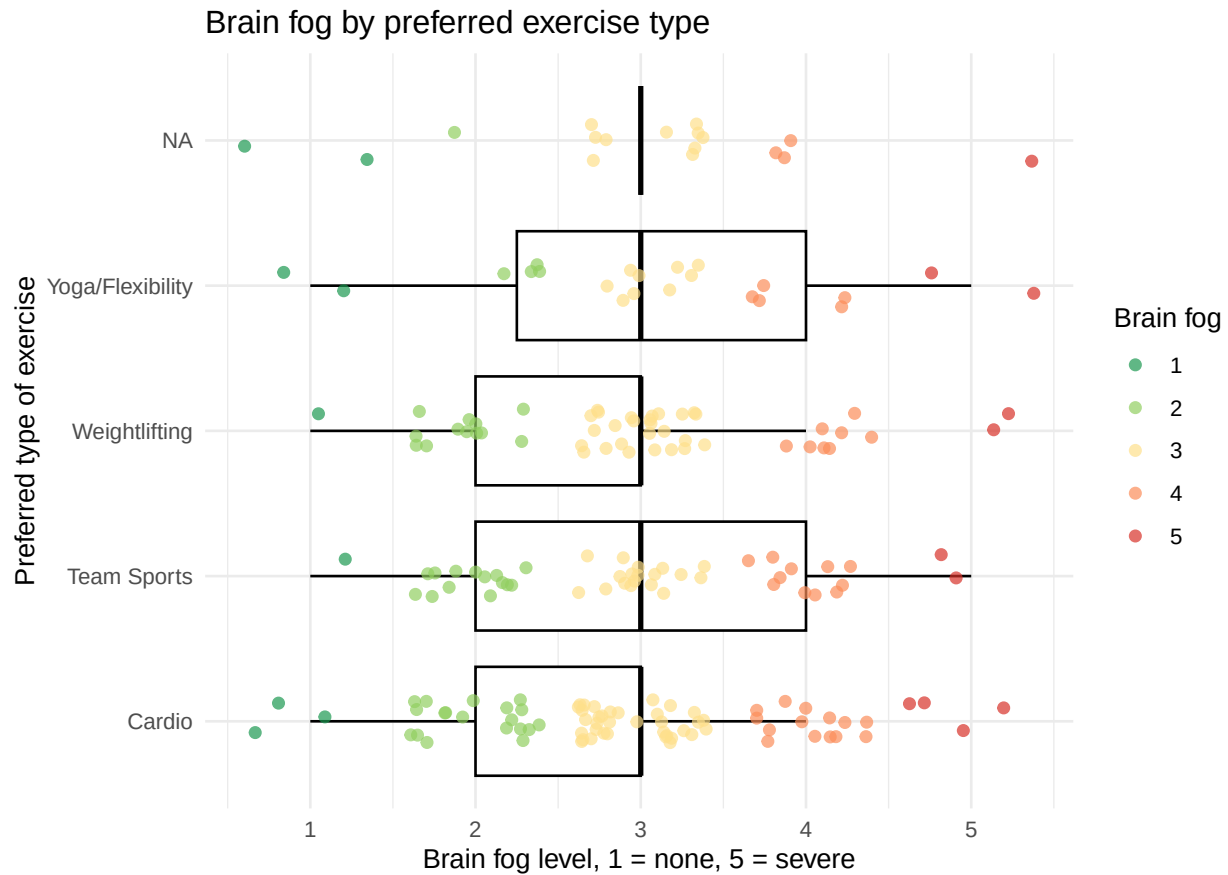
never exceed 200 minutes of weekly exercise, while those who exercise intensely (beyond 400 minutes) rarely report severe cognitive symptoms.

The hypothesis fails as a linear rule because high exercise does not automatically guarantee a clear mind.

RQ2: Brain fog by preferred exercise type

A boxplot is appropriate because the explanatory variable is categorical and the outcome variable is measured on a 1–5 scale.

```
ggplot(survey, aes(
  x = reorder(exercise_type, brain_fog, median, na.rm = TRUE),
  y = brain_fog,
  color = factor(brain_fog)
)) +
  geom_boxplot(color = "black", outlier.shape = NA) +
  geom_jitter(width = 0.15, alpha = 0.7, size = 1.8) +
  coord_flip() +
  scale_y_continuous(breaks = 1:5) +
  scale_color_manual(
    values = c(
      "1" = "#1a9850", # dark green: no brain fog
      "2" = "#91cf60", # light green
      "3" = "#fee08b", # yellow
      "4" = "#fc8d59", # orange
      "5" = "#d73027"  # red: severe brain fog
    ),
    name = "Brain fog"
  ) +
  labs(
    title = "Brain fog by preferred exercise type",
    x = "Preferred type of exercise",
    y = "Brain fog level, 1 = none, 5 = severe"
  ) +
  theme_minimal()
```

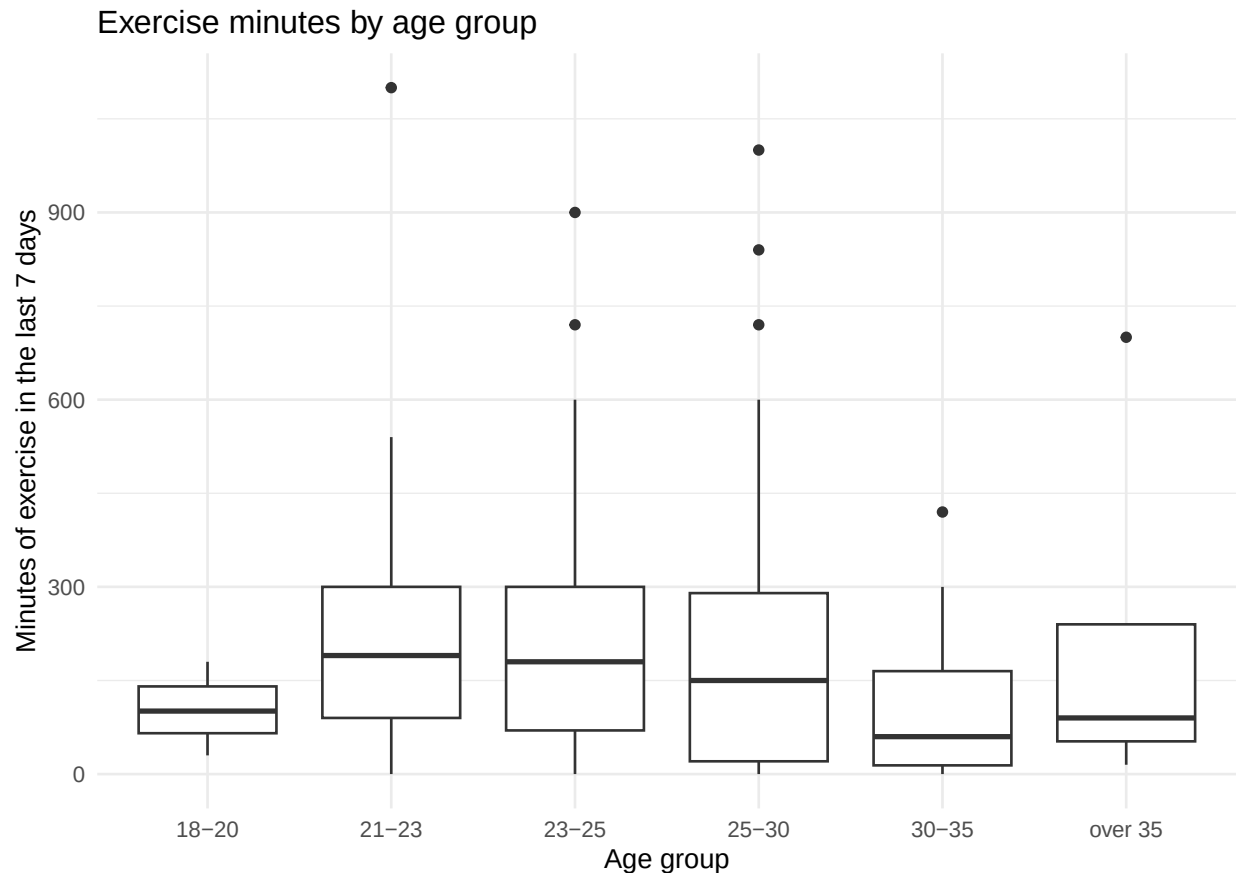


Finding: The lowest average brain fog is observed for Weightlifting, while the highest average brain fog is observed for Yoga/Flexibility. This comparison is descriptive and does not by itself prove that the type of exercise causes differences in brain fog.

RQ3: Age and exercise minutes

A scatterplot is used because both age and exercise minutes are quantitative variables. A trend line is added to show the general direction of the relationship.

```
ggplot(survey, aes(x = age, y = exercise_minutes)) +
  geom_boxplot() +
  labs(
    title = "Exercise minutes by age group",
    x = "Age group",
    y = "Minutes of exercise in the last 7 days"
  ) +
  theme_minimal()
```



Finding: The plot is used to assess whether older students tend to report fewer exercise minutes. The formal Spearman test below evaluates this negative-association hypothesis.

Descriptive inference

The following table summarizes the quantitative variables using sample size, mean, standard deviation, median, minimum, and maximum. The median is useful because exercise minutes may be right-skewed if some students report very high exercise times.

```
# One combined descriptive summary table

numeric_vars <- survey %>%
  select(exercise_minutes, brain_fog)

numeric_summary <- data.frame(
  Variable = c("Exercise minutes", "Brain fog"),
  Type = "Quantitative",
  Category = "",
  N = as.character(sapply(numeric_vars, function(x) sum(!is.na(x)))),
  Percent = "",
  Mean = as.character(round(sapply(numeric_vars, function(x) mean(x, na.rm = TRUE)), 2)),
  SD = as.character(round(sapply(numeric_vars, function(x) sd(x, na.rm = TRUE)), 2)),
  Median = as.character(round(sapply(numeric_vars, function(x) median(x, na.rm = TRUE)), 2)),
  Min = as.character(round(sapply(numeric_vars, function(x) min(x, na.rm = TRUE)), 2)),
  Max = as.character(round(sapply(numeric_vars, function(x) max(x, na.rm = TRUE)), 2))
)
```

```

)

categorical_table <- function(data, variable, label) {
  data %>%
    filter(!is.na(.data[[variable]])) %>%
    count(.data[[variable]], name = "N") %>%
    mutate(
      Variable = label,
      Type = "Categorical",
      Category = as.character(.data[[variable]]),
      Percent = paste0(round(100 * N / sum(N), 1), "%"),
      Mean = "",
      SD = "",
      Median = "",
      Min = "",
      Max = "",
      N = as.character(N)
    ) %>%
    select(Variable, Type, Category, N, Percent, Mean, SD, Median, Min, Max)
}

categorical_summary <- bind_rows(
  categorical_table(survey, "age", "Age group"),
  categorical_table(survey, "gender", "Gender"),
  categorical_table(survey, "academic_program", "Academic program"),
  categorical_table(survey, "exercise_type", "Preferred exercise type")
)

summary_table <- bind_rows(
  numeric_summary,
  categorical_summary
)

summary_table %>%
  knitr::kable(
    caption = "Descriptive summary statistics for Group 10 variables",
    booktabs = TRUE,
    align = "lllrrrrrrr"
  ) %>%
  kableExtra::kable_styling(
    latex_options = c("striped", "hold_position", "scale_down"),
    full_width = FALSE
  )
)

```

The next table summarizes the main categorical variable, preferred type of exercise.

```

exercise_type_summary <- survey %>%
  filter(!is.na(exercise_type)) %>%
  count(exercise_type, name = "n") %>%
  mutate(percent = round(100 * n / sum(n), 1)) %>%
  arrange(desc(n))

exercise_type_summary

```


Table 1: Descriptive summary statistics for Group 10 variables

Variable	Type	Category	N	Percent	Mean	SD	Median	Min	Max
Exercise minutes	Quantitative		209		195.2	195.43	150	0	1100
Brain fog	Quantitative		209		2.98	0.91	3	1	5
Age group	Categorical	18-20	3	1.4%					
Age group	Categorical	21-23	30	14.4%					
Age group	Categorical	23-25	55	26.3%					
Age group	Categorical	25-30	98	46.9%					
Age group	Categorical	30-35	15	7.2%					
Age group	Categorical	over 35	8	3.8%					
Gender	Categorical	Female	53	25.4%					
Gender	Categorical	Male	156	74.6%					
Academic program	Categorical	Business Informatics	32	15.3%					
Academic program	Categorical	Data Science	177	84.7%					
Preferred exercise type	Categorical	Cardio	75	39.1%					
Preferred exercise type	Categorical	Team Sports	46	24%					
Preferred exercise type	Categorical	Weightlifting	49	25.5%					
Preferred exercise type	Categorical	Yoga/Flexibility	22	11.5%					

```
## # A tibble: 4 x 3
##   exercise_type      n percent
##   <fct>          <int>   <dbl>
## 1 Cardio           75    39.1
## 2 Weightlifting     49    25.5
## 3 Team Sports       46     24
## 4 Yoga/Flexibility  22    11.5
```

Analytic inference

Because brain fog is measured on an ordinal 1–5 scale and exercise minutes may not be normally distributed, Spearman's rank correlation is suitable for RQ1 and RQ3. It tests whether two variables have a monotonic relationship. For RQ2, the Kruskal-Wallis test is used because it compares brain fog across multiple exercise-type groups without assuming normality.

```
test_results <- data.frame(
  Research_question = c("RQ1", "RQ2", "RQ3"),
  Test = c(
    "Spearman correlation",
    "Kruskal-Wallis rank-sum test",
    "Kruskal-Wallis rank-sum test"
  ),
  Hypothesis = c(
    "More exercise minutes are associated with lower brain fog",
    "Brain fog differs by preferred exercise type",
    "Exercise minutes differ by age group"
  ),
  Estimate = c(
    round(unname(rq1_test$estimate), 3),
    NA,
    NA
  ),
)
```

```

Statistic = c(
  round(unname(rq1_test$statistic), 2),
  round(unname(rq2_test$statistic), 2),
  round(unname(rq3_test$statistic), 2)
),
P_value = c(
  p_text(rq1_test$p.value),
  p_text(rq2_test$p.value),
  p_text(rq3_test$p.value)
),
Decision_5_percent = c(
  ifelse(rq1_test$p.value < 0.05, "Reject H0", "Do not reject H0"),
  ifelse(rq2_test$p.value < 0.05, "Reject H0", "Do not reject H0"),
  ifelse(rq3_test$p.value < 0.05, "Reject H0", "Do not reject H0")
)
)

test_results %>%
  rename(
    RQ = Research_question,
    `P value` = P_value,
    `Decision 5 percent` = Decision_5_percent
  ) %>%
  mutate(
    Hypothesis = kableExtra::linebreak(Hypothesis, align = "l")
  ) %>%
  knitr::kable(
    caption = "Analytic inference results",
    booktabs = TRUE,
    escape = FALSE,
    align = "lllrrll"
  ) %>%
  kableExtra::kable_styling(
    latex_options = c("striped", "hold_position"),
    full_width = FALSE
  ) %>%
  kableExtra::column_spec(1, width = "1cm") %>%
  kableExtra::column_spec(2, width = "3cm") %>%
  kableExtra::column_spec(3, width = "3cm")

```

Interpretation of analytic results

For **RQ1**: To address our first research question, we performed a Spearman rank correlation test to evaluate the relationship between weekly physical exercise and reported brain fog levels during lectures. This non-parametric test is given that our independent variable is an ordinal scale from 1 to 5, which violates the normality assumptions required for standard linear models.

The test yielded a Spearman correlation coefficient estimate of -0.113 , indicating a very weak negative relationship between the two variables. The resulting p-value was calculated to be 0.051 . Because this p-value is slightly above our strict 5% significance threshold ($\alpha = 0.05$), we do not reject the null hypothesis (H_0), which states that there is no monotonic association between exercise minutes and brain fog.

Table 2: Analytic inference results

RQ	Test	Hypothesis	Estimate	Statistic	P value	Decision 5 percent
RQ1	Spearman correlation	More exercise minutes are associated with lower brain fog	-0.113	1694191.23	0.051	Do not reject H0
RQ2	Kruskal-Wallis rank-sum test	Brain fog differs by preferred exercise type	NA	0.24	0.972	Do not reject H0
RQ3	Kruskal-Wallis rank-sum test	Exercise minutes differ by age group	NA	6.31	0.278	Do not reject H0

The statistical test confirms that we cannot definitively claim a general trend of lower brain fog for students who exercise more across the entire dataset. While our visualization hinted at a protective threshold at the extreme ends, specifically that severe brain fog is isolated to lower exercise levels, the weak correlation coefficient and the borderline non significant p-value demonstrate that this pattern is not strong or consistent enough across the central categories to be statistically reliable.

Ultimately, these findings suggest that while high levels of physical activity may limit severe cognitive fatigue for some, minor fluctuations in weekly exercise minutes do not systematically predict lower levels of day to day brain fog for the broader student body.

For **RQ2**, the Kruskal-Wallis test has a p-value of 0.972. At the 5% significance level, we do not reject the null hypothesis, so the data do not provide strong evidence that brain fog differs by preferred exercise type.

For **RQ3**, the Kruskal-Wallis test has a p-value of `r p_text(rq3_testp.value)`. At the 5% significance level, `p.value < 0.05`, “reject the null hypothesis and conclude that weekly exercise minutes differ between at least some age groups”, “do not reject the null hypothesis, so the data do not provide strong evidence that weekly exercise minutes differ between age groups”).

Functions used

Function	Purpose in this report
<code>read_csv()</code>	Loads the CSV survey file.
<code>select()</code> and <code>starts_with()</code>	Extract the demographic and Group 10 variables.
<code>str_remove()</code>	Removes the prefixes DQ and G10 from column names.
<code>rename()</code>	Gives the selected columns shorter names.
<code>mutate()</code>	Converts variables and creates analysis-ready values.
<code>filter()</code>	Removes missing values before plots, summaries, and tests.
<code>ggplot()</code>	Creates all visualizations.
<code>geom_boxplot()</code> , <code>geom_jitter()</code> , <code>geom_point()</code> , <code>geom_smooth()</code>	Show distributions, individual observations, scatterplots, and trend lines.
<code>count()</code> and <code>summarise()</code>	Compute categorical frequencies and summary statistics.
<code>cor.test()</code>	Performs Spearman correlation tests for RQ1 and RQ3.
<code>kruskal.test()</code>	Tests whether brain fog differs across exercise-type groups for RQ2.
<code>knitr::kable()</code>	Displays results in readable tables.

Function	Purpose in this report
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Conclusion

This report explored the relationship between physical exercise habits and mental brain fog among students. The descriptive analysis summarizes how much students exercise, which exercise types they prefer, and how strongly they report brain fog during lectures. The exploratory plots show the observed patterns visually, while the analytic tests evaluate whether these patterns are statistically supported.

Overall, the findings should be interpreted as associations rather than causal effects. The survey is observational, so it cannot prove that exercise directly causes changes in brain fog. However, the results provide useful evidence about whether students with different exercise habits also report different levels of focus-related difficulty during lectures.